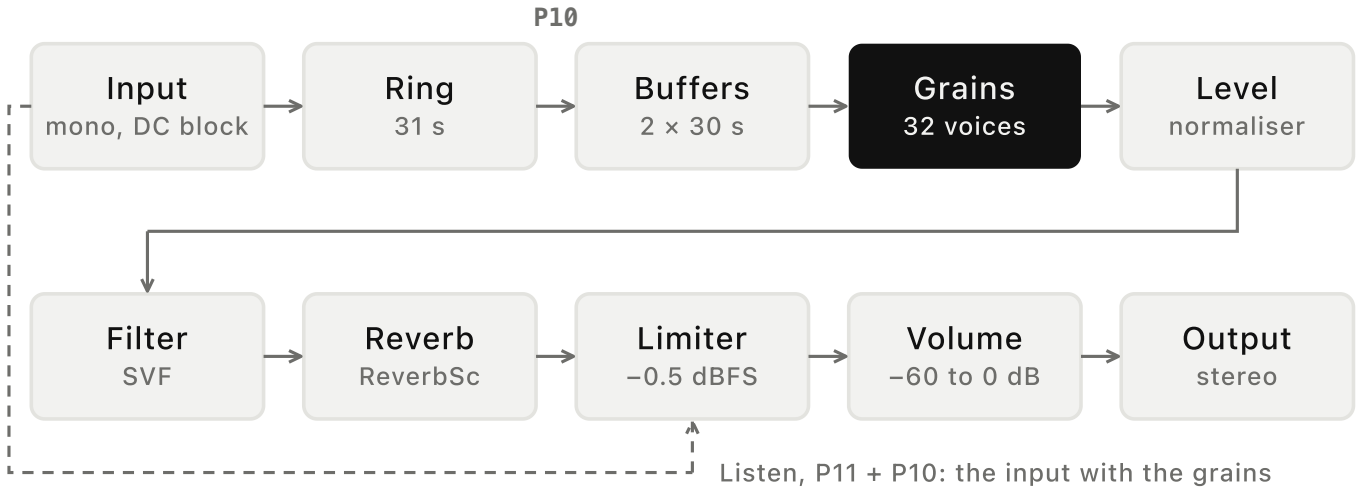
Engine

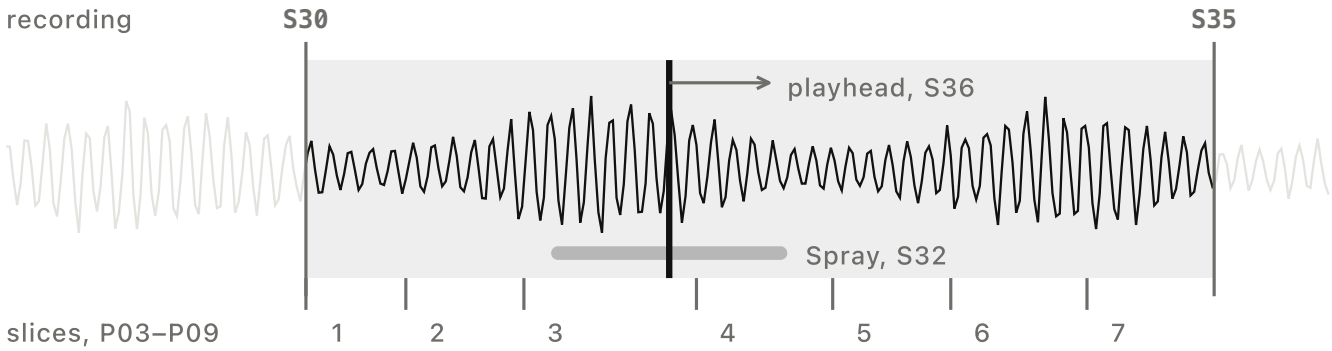
How a recording becomes grains, and how they fall.

01 SIGNAL PATH



Both jacks are summed to mono, −3 dB each. The ring takes the input all the time. P10 copies from it into the buffer not playing, and the grains read the buffer playing.

02 RECORDING



TAP	Under 0.3 s: the 8 s before it
HOLD	From 50 ms before the touch to the release, up to 30 s
TRIM	Silence under −60 dBFS cut from both ends
LAND	Copied at 8× real time. Over 2 s each new grain reads it with rising odds, the rest the recording before. The first rises from silence
SLICES	Its attacks found once, in 5 ms frames, the strongest 64 kept. A loop's seven slices: its start, then its strongest attacks. Too few, as a held note: even parts

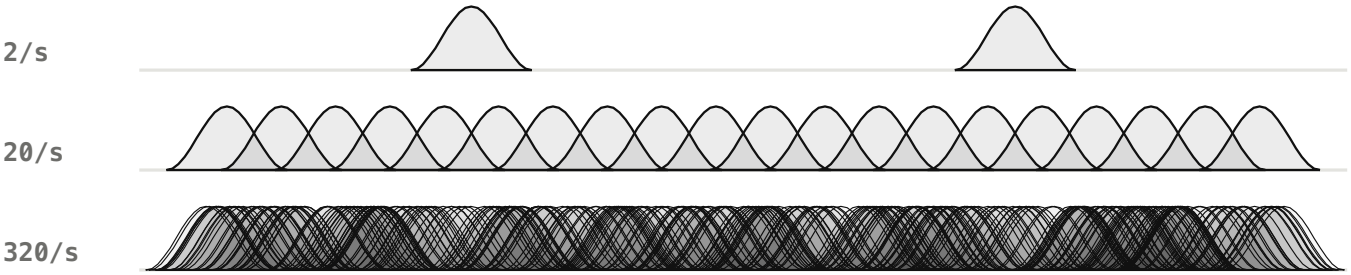
03 A GRAIN

POSITION	The playhead, and with Spray anywhere round it, evenly, up to the whole loop
START	Moved, at most 5.3 ms or a quarter of its length, to where it continues the grain before in phase: the signs of 512 samples compared, as Clouds' WSOLA player. Not with Spray up, whose grains are meant to differ
READ	Four-point Hermite between samples, as Beads. Rate 2^(n/12): Pitch ±24 semitones, stops at 0, 1, 3, 4, 7, 12, 24
RANDOM PITCH	Up: any pitch to ±12 semitones. Down: octaves, up to two grains in three
SIZE	5 ms to 2 s, 100 ms at the centre. Random size: part of the way to any size, exponential
WINDOW	Beads' envelope along Shape: square, decay, bell, reverse. At least 0.5 ms up and 10 ms down
PAN	Spray spreads them, wider as more sound at once. Equal power
DIRECTION	The playhead's. Reverse sets the share read the other way



square decay bell reverse

04 STARTS



Density is grains a second, 0.25 to 320, none at the bottom, whatever their size: grains sounding at once are Density × Size. Random timing takes the starts from a regular pulse to nearly Poisson, the same mean. 32 voices: when all sound, a new grain is dropped.

05 LEVEL



in phase
amplitudes add

scattered
powers add

Grains in phase add in amplitude, scattered ones in power. A normaliser measures both every block, 5 ms smoothed, and brings the sum to the input's level with 0.9 headroom, never boosting grains that cancel. Aligned starts keep slowed grains in phase: at ½× a sawtooth's harmonics vary 0.27 dB from one 20 ms frame to the next, as at 1×. Without, 1.0 dB, and 5.6 dB in a dense random cloud.

06 TRANSPORT

SPEED	0 to 4×, stops at ¼, ½, 1, 2, 4. The pitch stays. Frozen by S09, the fader scrubs: a full stroke the whole loop
END	At the loop's end, S07: once and stop, turn over and run back, or round again
STOP	P00 held: everything slows to a halt over 0.6 s, as a tape machine, faded over its slowest quarter. Released: back to speed
TRIGGER	P02: the loop from its start (its end in reverse), a grain at once. After Once, it pours again
SLICES	P03–P09 held: each plays its slice with its own playhead, at Speed and Density, the loop waiting. Several at once

07 EFFECTS

FILTER	SVF on the grains. Low-pass 20 kHz to 200 Hz, off at the centre, high-pass 20 Hz to 2 kHz. Resonance 0.7 to 1.5
REVERB	ReverbSc on the grains. Decay 0.5 to 16 s, damped at 10 kHz, up to equal parts with the dry grains
LIMITER	Look-ahead 5 ms, −0.5 dBFS. The grains +6 dB into it
VOLUME	After the limiter: off, then −60 to 0 dB

08 RUNTIME

AUDIO	48 kHz, 48-sample blocks: 1 ms. Recordings in SDRAM, mono
CPU	15 % with no grains. 83 % at the most: 32 voices, all slices held, Pitch +24, a recording landing
BROWSER	The same C++ core in WebAssembly, in an AudioWorklet